

Pregnancy Risk Model Testing

Real Test: Paramètres_original_extended.xlsx

Real rows used: 100

BS conversion applied during testing: $BS_real(g/L) * 10$

Risk mapping: 0 -> LOW, + -> MEDIUM, ++ -> HIGH

Compared models: Random Forest, Logistic Regression, SVM, XGBoost

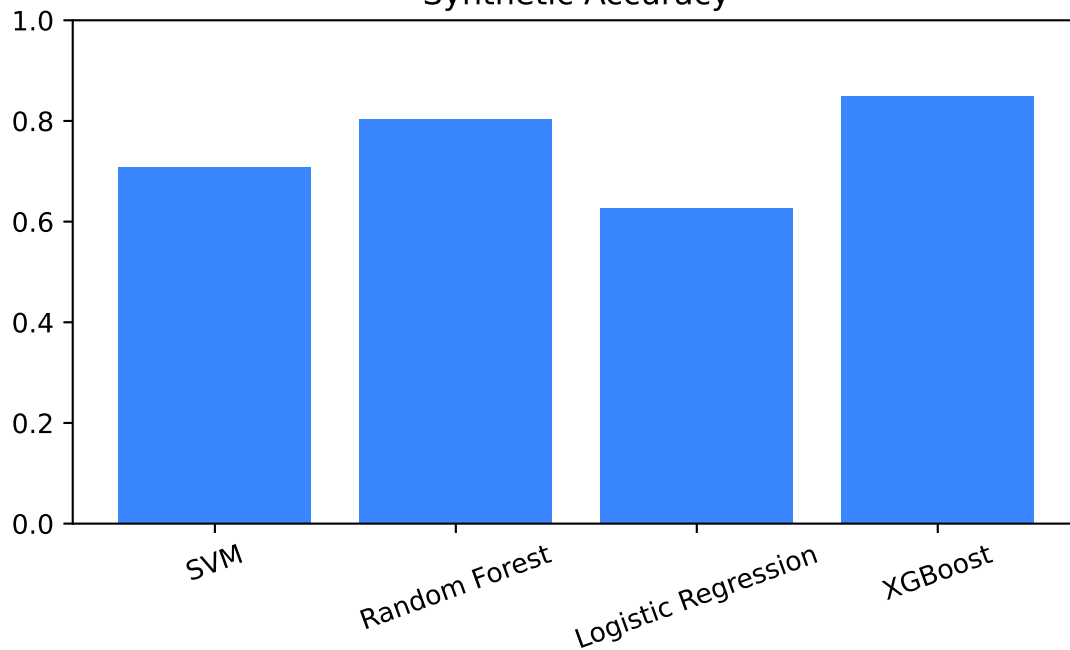
Metrics: synthetic accuracy, real accuracy, macro F1, F1 per class

Performance Summary

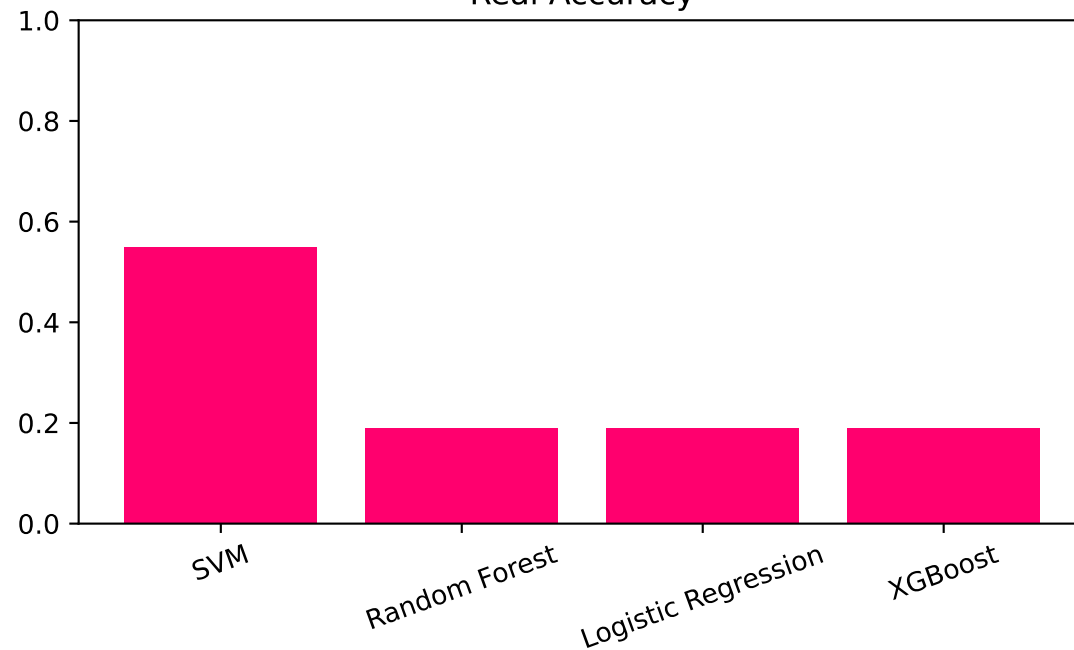
Model	Train(s)	Syn Acc	Real Acc	Real MacroF1	F1 HIGH	F1 LOW	F1 MEDIUM
SVM	0.098	0.708	0.550	0.237	0.000	0.710	0.000
Random Forest	0.485	0.803	0.190	0.106	0.319	0.000	0.000
Logistic Regression	0.024	0.626	0.190	0.106	0.319	0.000	0.000
XGBoost	0.305	0.849	0.190	0.106	0.319	0.000	0.000

Model Dashboard

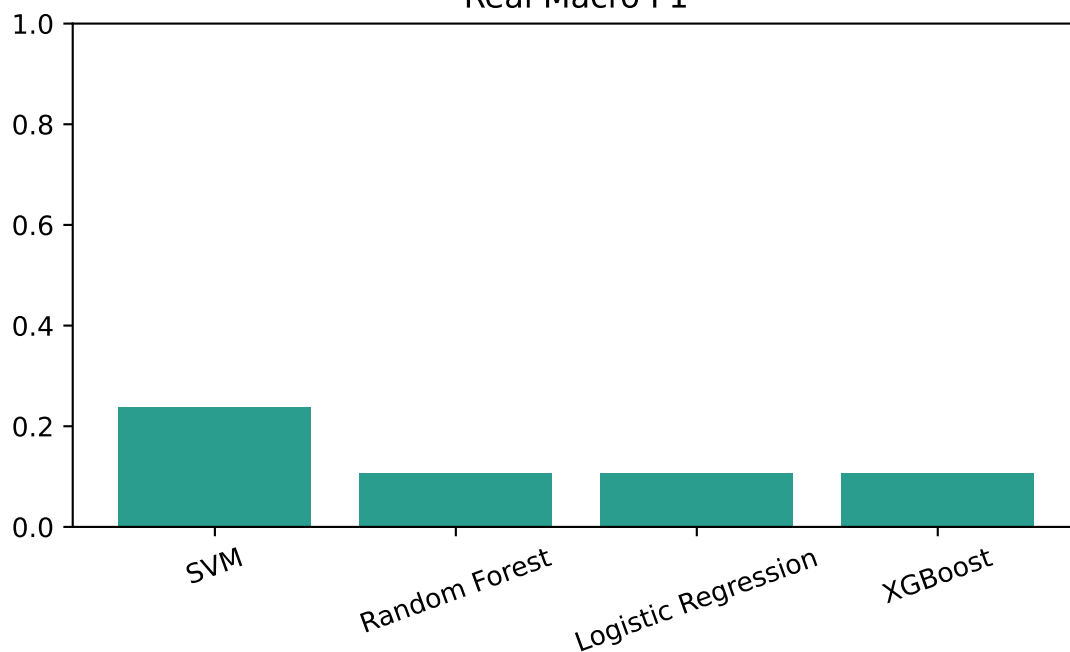
Synthetic Accuracy



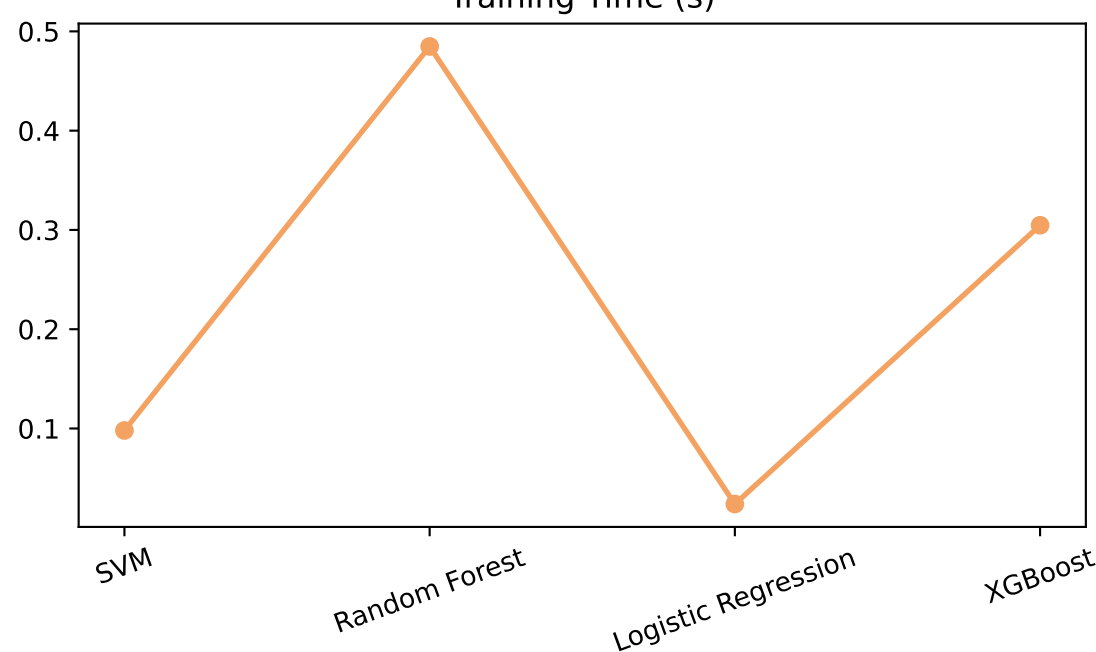
Real Accuracy



Real Macro F1



Training Time (s)



Strengths, Weaknesses, and Test Points

SVM

Power: Best F1 on LOW: 0.710 | Best recall on LOW: 1.000 | Best precision on LOW: 0.550

Weak: Weakest F1 on HIGH: 0.000 | Weakest recall on HIGH: 0.000 | Weakest precision on HIGH: 0.000

Good test points: idx 0 LOW->LOW conf=0.43 ; idx 1 LOW->LOW conf=0.43 ; idx 2 LOW->LOW conf=0.43

Not good test points: idx 8 MEDIUM->LOW conf=0.43 ; idx 9 MEDIUM->LOW conf=0.43 ; idx 10 HIGH->LOW conf=0.43

Random Forest

Power: Best F1 on HIGH: 0.319 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.190

Weak: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

Good test points: idx 49 HIGH->HIGH conf=1.00 ; idx 50 HIGH->HIGH conf=1.00 ; idx 10 HIGH->HIGH conf=0.99

Not good test points: idx 24 MEDIUM->HIGH conf=0.99 ; idx 17 MEDIUM->HIGH conf=0.99 ; idx 23 MEDIUM->HIGH conf=0.99

Logistic Regression

Power: Best F1 on HIGH: 0.319 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.190

Weak: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

Good test points: idx 10 HIGH->HIGH conf=1.00 ; idx 16 HIGH->HIGH conf=1.00 ; idx 22 HIGH->HIGH conf=1.00

Not good test points: idx 8 MEDIUM->HIGH conf=1.00 ; idx 13 MEDIUM->HIGH conf=1.00 ; idx 14 MEDIUM->HIGH conf=1.00

XGBoost

Power: Best F1 on HIGH: 0.319 | Best recall on HIGH: 1.000 | Best precision on HIGH: 0.190

Weak: Weakest F1 on LOW: 0.000 | Weakest recall on LOW: 0.000 | Weakest precision on LOW: 0.000

Good test points: idx 49 HIGH->HIGH conf=1.00 ; idx 50 HIGH->HIGH conf=1.00 ; idx 22 HIGH->HIGH conf=1.00

Not good test points: idx 23 MEDIUM->HIGH conf=1.00 ; idx 24 MEDIUM->HIGH conf=1.00 ; idx 45 MEDIUM->HIGH conf=1.00